

```

BEAN-
package com.accenture.lkm.business.bean;

import java.util.Date;

public class BillBean {
    private Integer billNo;
    private String patientName;
    private Integer testId;
    private Date dateOfCollection;
    private Double paidAmount;

    private String testName;
    private Double balanceAmount;

    public String getTestName() {
        return testName;
    }
    public void setTestName(String testName) {
        this.testName = testName;
    }
    public Double getBalanceAmount() {
        return balanceAmount;
    }
    public void setBalanceAmount(Double balanceAmount) {
        this.balanceAmount = balanceAmount;
    }
    public Integer getBillNo() {
        return billNo;
    }
    public void setBillNo(Integer billNo) {
        this.billNo = billNo;
    }
    public String getPatientName() {
        return patientName;
    }
    public void setPatientName(String patientName) {
        this.patientName = patientName;
    }
    public Integer getTestId() {
        return testId;
    }
    public void setTestId(Integer testId) {
        this.testId = testId;
    }
    public Date getDateOfCollection() {
        return dateOfCollection;
    }
    public void setDateOfCollection(Date dateOfCollection) {
        this.dateOfCollection = dateOfCollection;
    }
    public Double getPaidAmount() {
        return paidAmount;
    }
    public void setPaidAmount(Double paidAmount) {
        this.paidAmount = paidAmount;
    }
    @Override
    public String toString() {

```

```
        return "BillBean [billNo=" + billNo + ", patientName=" +
patientName + ", testId=" + testId
                + ", dateOfCollection=" + dateOfCollection + ",
paidAmount=" + paidAmount + "]" ;
    }
```

```
}
```

```
-----
```

```
package com.accenture.lkm.business.bean;
```

```
import java.util.Date;
```

```
public class DateRangeBean {
```

```
    private Date from;
    private Date to;
    public Date getFrom() {
        return from;
    }
    public void setFrom(Date from) {
        this.from = from;
    }
    public Date getTo() {
        return to;
    }
    public void setTo(Date to) {
        this.to = to;
    }
}
```

```
}
```

```
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```

```
package com.accenture.lkm.business.bean;
```

```
public class TestBean {
```

```
    private Integer testId;
    private String testName;
    private Double charges;
    public Integer getTestId() {
        return testId;
    }
    public void setTestId(Integer testId) {
        this.testId = testId;
    }
    public String getTestName() {
        return testName;
    }
    public void setTestName(String testName) {
        this.testName = testName;
    }
    public Double getCharges() {
        return charges;
    }
    public void setCharges(Double charges) {
        this.charges = charges;
    }
}
```

```

    }
    @Override
    public String toString() {
        return "TestBean [testId=" + testId + ", testName=" + testName
+ ", charges=" + charges + "]\n";
    }
}

```

```

}
-
=====
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```

DAo-

```
package com.accenture.lkm.dao;
```

```
import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.RepositoryDefinition;
import org.springframework.transaction.annotation.Transactional;
```

```
import com.accenture.lkm.entity.BillEntity;
```

```
@RepositoryDefinition(domainClass = BillEntity.class, idClass =
Integer.class)
```

```
@Transactional(value = "txManager")
```

```
public interface BillDAO {
    @Query(name = "save")
    BillEntity save(BillEntity entity);
}

```

```
-----
package com.accenture.lkm.dao;
```

```
import java.util.Date;
import java.util.List;
```

```
import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.RepositoryDefinition;
import org.springframework.transaction.annotation.Transactional;
```

```
import com.accenture.lkm.entity.TestEntity;
```

```
@RepositoryDefinition(idClass = Integer.class, domainClass =
TestEntity.class)
```

```
@Transactional(value = "txManager")
```

```
// name of TransactionManger can be any thing
```

```
public interface TestDAO {
    @Query(name = "findAllTestDetails")
    List<TestEntity> findAllTestDetails();
}

```

```
-----
----
package com.accenture.lkm.dao;
```

```
import java.util.ArrayList;
import java.util.Date;
import java.util.List;
```

```
import javax.persistence.EntityManager;
import javax.persistence.PersistenceContext;
import javax.persistence.Query;
```

```

import org.springframework.beans.BeanUtils;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;

import com.accenture.lkm.business.bean.BillBean;
import com.accenture.lkm.business.bean.TestBean;
import com.accenture.lkm.entity.BillEntity;
import com.accenture.lkm.entity.TestEntity;

@Repository
@Transactional(value = "txManager")
public class TestDAOWrapper {

    @Autowired
    private TestDAO testDAO;
    @Autowired
    private BillDAO billDAO;

    @PersistenceContext
    private EntityManager entityManager;

    // Utility Methods.....
    public static BillBean convertBillEntityToBean(BillEntity entity) {
        BillBean bean = new BillBean();
        BeanUtils.copyProperties(entity, bean);
        return bean;
    }
    public static BillEntity convertBillBeanToEntity(BillBean bean) {
        BillEntity entity = new BillEntity();
        BeanUtils.copyProperties(bean, entity);
        return entity;
    }
    public static TestBean convertTestEntityToBean(TestEntity entity) {
        TestBean customerBean = new TestBean();
        BeanUtils.copyProperties(entity, customerBean);
        return customerBean;
    }
    public static TestEntity convertTestBeanToEntity(TestBean bean) {
        TestEntity entity = new TestEntity();
        BeanUtils.copyProperties(bean, entity);
        return entity;
    }

    public List<TestBean> findAllTest() {
        List<TestEntity> entityList = testDAO.findAllTestDetails();
        List<TestBean> beanList = null;
        if (entityList != null) {
            beanList = new ArrayList<TestBean>();
            for (TestEntity entity: entityList) {
                beanList.add(convertTestEntityToBean(entity));
            }
        }

        return beanList;
    }
}

```

```

        public BillBean addBillDetails(BillBean billBean) {
            BillEntity billEntity = convertBillBeanToEntity(billBean);
            entityManager.persist(billEntity);
//            billDAO.save(billEntity);
            return convertBillEntityToBean(billEntity);
        }

        public TestBean getTestById(Integer testId) {
            TestEntity testEntity = entityManager.find(TestEntity.class,
testId);
            return convertTestEntityToBean(testEntity);
        }

        public List<BillBean> getTestBillDetails(Date from, Date to) {
            Query query = entityManager.createQuery("select k from
BillEntity k where k.dateOfCollection between ?1 and ?2");
            query.setParameter(1, from);
            query.setParameter(2, to);

            List<BillBean> beanList = null;
            List<BillEntity> entityList = query.getResultList();
            if (entityList != null) {
                beanList = new ArrayList<BillBean>();
                for (BillEntity entity:entityList) {
                    beanList.add(convertBillEntityToBean(entity));
                }
            }
            return beanList;
        }
    }
}

```

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```

```

Entity-
package com.accenture.lkm.entity;

import java.util.Date;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
import javax.persistence.Temporal;
import javax.persistence.TemporalType;

import org.hibernate.validator.constraints.NotEmpty;

@Entity
@Table(name = "Bill")
public class BillEntity {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Integer billNo;
    @NotEmpty
    private String patientName;

```

```

@Column(name = "testID")
private Integer testId;
@Temporal(TemporalType.DATE)
private Date dateOfCollection;
private Double paidAmount;
public Integer getBillNo() {
    return billNo;
}
public void setBillNo(Integer billNo) {
    this.billNo = billNo;
}
public String getPatientName() {
    return patientName;
}
public void setPatientName(String patientName) {
    this.patientName = patientName;
}
public Integer getTestId() {
    return testId;
}
public void setTestId(Integer testId) {
    this.testId = testId;
}
public Date getDateOfCollection() {
    return dateOfCollection;
}
public void setDateOfCollection(Date dateOfCollection) {
    this.dateOfCollection = dateOfCollection;
}
public Double getPaidAmount() {
    return paidAmount;
}
public void setPaidAmount(Double paidAmount) {
    this.paidAmount = paidAmount;
}
@Override
public String toString() {
    return "BillEntity [billNo=" + billNo + ", patientName=" +
patientName + ", testId=" + testId
        + ", dateOfCollection=" + dateOfCollection + ",
paidAmount=" + paidAmount + "];"
}
}

```

```

}
-----
package com.accenture.lkm.entity;

```

```

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.Id;
import javax.persistence.Table;

```

```

@Entity
@Table(name = "Test")
public class TestEntity {

```

```

    @Id
    @Column(name = "testID")
    private Integer testId;

```

```

private String testName;
private Double charges;
public Integer getTestId() {
    return testId;
}
public void setTestId(Integer testId) {
    this.testId = testId;
}
public String getTestName() {
    return testName;
}
public void setTestName(String testName) {
    this.testName = testName;
}
public Double getCharges() {
    return charges;
}
public void setCharges(Double charges) {
    this.charges = charges;
}
@Override
public String toString() {
    return "TestEntity [testId=" + testId + ", testName=" +
testName + ", charges=" + charges + "]";
}
}

```

```

}

```

```

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```

```

Resource-
create database if not exists springmvcmock;
use springmvcmock;

```

```

create table if not exists Test(
testID int primary key not null,
testName varchar(35),
charges double);

```

```

create table if not exists Bill(
billNo int primary key auto_increment,
patientName varchar(35) not null,
testID int references Test(testID),
dateOfCollection date,
paidAmount double
);

```

```

insert into Test(testId, testName, charges) values
(1001, "X-RAY REPORT", 200),
(1002, "ULTRASONOGRAPHY", 1200),
(1003, "ECG REPORT", 1000),
(1004, "EEG REPORT", 3000);

```

```

insert into Bill(billNo, patientName, testID, dateOfCollection,
paidAmount) values
(5001, "Harry", 1003, "2020-06-01", 800),
(5002, "John", 1004, "2020-05-16", 200),
(5003, "Madam", 1001, "2020-04-08", 150);

```

```

commit;
=====
=====
Service-
package com.accenture.lkm.service;

import java.util.Date;
import java.util.List;
import java.util.Map;

import com.accenture.lkm.business.bean.BillBean;
import com.accenture.lkm.business.bean.TestBean;

public interface TestService {
    List<TestBean> findAllTest();
    BillBean addBillDetails(BillBean billBean);
    TestBean getTestById(Integer testId);
    List<BillBean> getTestBillDetails(Date from, Date to);

    Map<Integer, String> populateTestMap();
}
-----
package com.accenture.lkm.service;

import java.util.Date;
import java.util.HashMap;
import java.util.List;
import java.util.Map;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.accenture.lkm.business.bean.BillBean;
import com.accenture.lkm.business.bean.TestBean;
import com.accenture.lkm.dao.TestDAOWrapper;

@Service
public class TestServiceImpl implements TestService{
    @Autowired
    private TestDAOWrapper testDAOWrapper;

    @Override
    public List<TestBean> findAllTest() {
        // TODO Auto-generated method stub
        return testDAOWrapper.findAllTest();
    }

    @Override
    public BillBean addBillDetails(BillBean billBean) {
        // TODO Auto-generated method stub
        return testDAOWrapper.addBillDetails(billBean);
    }

    @Override
    public TestBean getTestById(Integer testId) {
        // TODO Auto-generated method stub
        return testDAOWrapper.getTestById(testId);
    }
}

```

```

@Override
public List<BillBean> getTestBillDetails(Date from, Date to) {
    // TODO Auto-generated method stub
    return testDAOWrapper.getTestBillDetails(from, to);
}

@Override
public Map<Integer, String> populateTestMap() {
    List<TestBean> testList = findAllTest();
    Map<Integer, String> testMap = new HashMap<Integer, String>();
    for(TestBean bean:testList) {
        testMap.put(bean.getTestId(), bean.getTestName());
    }
    return testMap;
}
}

=====
=====
CONTROLLER-
package com.accenture.lkm.web.controller;

import java.util.HashMap;
import java.util.List;
import java.util.Map;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.servlet.ModelAndView;

import com.accenture.lkm.business.bean.BillBean;
import com.accenture.lkm.business.bean.DateRangeBean;
import com.accenture.lkm.business.bean.TestBean;
import com.accenture.lkm.service.TestService;

@Controller
public class TestController {

    @Autowired
    private TestService testService;

    @RequestMapping("/loadAddBillDetailsPage")
    public ModelAndView loadAddBillDetailsPage() {
        return new ModelAndView("AddBill", "billBean", new
BillBean());
    }

    @RequestMapping("/addBillDetails")
    public ModelAndView
addBillDetails(@ModelAttribute("billBean")BillBean billBean) {
        ModelAndView mv = new ModelAndView();

        billBean.setTestName(testService.getTestById(billBean.getTestId()).
getTestName());

```

```

        billBean.setBalanceAmount(testService.getTestById(billBean.getTestId()).getCharges()-billBean.getPaidAmount());
        BillBean bean = testService.addBillDetails(billBean);
        mv.setViewName("AddBillSuccess");
        mv.addObject("message", "Added bill details successfully with bill number: "+bean.getBillNo()+" . Balance amount to be paid="+billBean.getBalanceAmount());
        return mv;
    }

    @RequestMapping("/loadGetBillDetailsPage")
    public ModelAndView getBillDetailsPage() {
        return new ModelAndView("GetBill", "dateRangeBean", new DateRangeBean());
    }

    @RequestMapping("/getBillDetails")
    public ModelAndView getBillDetails(@ModelAttribute("dateRangeBean") DateRangeBean dateRangeBean) {
        List<BillBean> billList = testService.getTestBillDetails(dateRangeBean.getFrom(), dateRangeBean.getTo());

        ModelAndView mv = new ModelAndView();
        mv.setViewName("GetBill");
        mv.addObject("billList", billList);
        return mv;
    }

    @ModelAttribute("testMap")
    public Map<Integer, String> populateTestMap() {
        // List<TestBean> testList = testService.findAllTest();
        // Map<Integer, String> testMap = new HashMap<Integer, String>();
        // for(TestBean bean:testList) {
        //     testMap.put(bean.getTestId(), bean.getTestName());
        // }
        Map<Integer, String> testMap = testService.populateTestMap();
        return testMap;
    }

    //Error Handler:
    @ExceptionHandler(value=Exception.class)
    public ModelAndView handleAllExceptions(Exception exception){
        ModelAndView modelAndView = new ModelAndView();
        modelAndView.setViewName("GeneralizedExceptionHandlerPage");
        modelAndView.addObject("message", exception.getMessage());
        modelAndView.addObject("exception", exception);
        return modelAndView;
    }
}

```

```

}
=====
JSP VIEWS-
ADDBILL-
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>

```

```

<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<%@ taglib uri="http://www.springframework.org/tags" prefix="spring"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>Add Bill Page</title>
<style>
.error {
    color: #ff0000;
    font-style: italic;
}
</style>
</head>
<body>
    <center>
        <h2>Add Bill Details</h2>
        <form:form modelAttribute="billBean"
action="addBillDetails.html">
            <table border="2">
                <tr>
                    <td>Patient Name</td>
                    <td><form:input path="patientName" /></td>
                </tr>
                <tr>
                    <td>Test</td>
                    <td><form:radiobuttons path="testId"
items="${testMap}" /></td>
                </tr>
                <tr>
                    <td>Date of collection</td>
                    <td><form:input path="dateOfCollection" />
                </tr>
                <tr>
                    <td>Paid Amount</td>
                    <td><form:input path="paidAmount" />
                </tr>
            </table>
            <br>
            <input type="submit" value="Add" />
        </form:form>
        <br>
        <br>
        <a href="${pageContext.request.contextPath}/index.jsp">Home</a>
    </center>
</body>
</html>

```

ADBILLSUCCESS-

```

<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<%@ taglib uri="http://www.springframework.org/tags" prefix="spring"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>

```

```

<meta charset="ISO-8859-1">
<title>Add Bill Success</title>
</head>
<body>
    <center>
        <h2>Bill Added Successfully!</h2>
        <br>
        <br>
        <h3>${message}</h3>
        <br>
        <br>
        <a href="${pageContext.request.contextPath}/index.jsp">Home</a>
    </center>
</body>
</html>

```

```

-----
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<%@ taglib uri="http://www.springframework.org/tags" prefix="spring"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta charset="ISO-8859-1">
<title>Get Bill Details</title>
</head>
<body>
<center>
<h2>Get Bill Details Within Date Range</h2>
    <form:form action="getBillDetails.html"
modelAttribute="dateRangeBean">
        <table border="2">
            <tr>
                <td>From date</td>
                <td><form:input path="from"/></td>
            </tr>
            <tr>
                <td>To date</td>
                <td><form:input path="to"/></td>
            </tr>
        </table>
        <br>
        <input type="submit" value="Fetch Details"/>
    </form:form>
    <br>
    <c:if test="${empty billList}">
        <h3>No Bills in the given date range!!!</h3>
    </c:if>
    <c:if test="${not empty billList}">
        <table border="2">
            <tr>
                <td>Bill No</td>
                <td>Patient Name</td>
                <td>Test Id</td>
                <td>Date of Collection</td>
                <td>Paid Amount</td>
                <!-- <td>Test Name</td>

```

```

        <td>Balance Amount</td> -->
    </tr>
    <c:forEach var="bill" items="${billList}">

        <tr>
            <td>${bill.billNo}</td>
            <td>${bill.patientName}</td>
            <td>${bill.testId}</td>
            <td>${bill.dateOfCollection}</td>
            <td>${bill.paidAmount}</td>
            <!-- <td>${bill.testName}</td> -->
            <!-- <td>${bill.balanceAmount}</td> -->
        </tr>

    </c:forEach>
</table>
</c:if>
<br>
<br>
<a href="${pageContext.request.contextPath}/index.jsp">Home</a>
</center>
</body>
</html>

```

```

=====
=====
INDEX-
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<center>
    <h2>Wellness Diagnostic Center</h2>
    <br>
    <br>
    <a
href="${pageContext.request.contextPath}/loadAddBillDetailsPage.html">Add
Bill</a>
    <br>
    <br>
    <a
href="${pageContext.request.contextPath}/loadGetBillDetailsPage.html">Get
Bill Details</a>

</center>
</body>
</html>
=====
=====

```